

From Obstruction Certificates to Monitoring Standards: An Applications and Limitations Statement for the Obstruction Calculus

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Abstract

This statement consolidates, in one place and in plain scope, what the obstruction calculus and its follow-up programme are for, whom they serve today, and what they do not yet do. It is a satellite document: it creates no new theorems and modifies no manuscript of the programme; every claim below is a pointer into a cited source with a machine-verifiable record. Four design rules are extracted in usable form — review-interval sizing, the fibre rule for indicator aggregation, bias correction, and structural-cause forensics — each with its scope, its source theorem, and its verified instance. The operational-versus-architectural distinction is stated explicitly, a stakeholder scope table is given, and the limitations are recorded with the same precision as the capabilities: model-availability, the dimensionality wall of exact recursions, the deferred numerical campaign, the absence of calibrated case studies, and the verification-grade status of the software. The programme’s delivered lineages are indexed against the utility prerequisites, and the genuinely open items are listed.

1. Purpose and status

The obstruction calculus (Abaee, 2026a) proves, from a model of a controlled system and its observation structure, that certain information deficits — coarse aggregation, delayed review, biased reading, unavailable certification — make robust viability impossible even when every physical state is viable in isolation. The follow-up programme has carried the calculus to a continuous-to-finite bridge with solver-level certificates (Abaee, 2026b), a selector unification (Abaee, 2026c), an exact stochastic belief-state value theory (Abaee, 2026d), a five-layer successor architecture (Abaee, 2026e), and a generalization catalogue of ten further lineages (Abaee, 2026f). This statement is a satellite of the whole set: it translates the results into the form in which they can be used — and states, with the same care, the boundary of that use. Nothing here creates theorem status; every pointer resolves to a source with an exact-arithmetic verification record, and the pointers themselves are machine-checked (Section 7).

2. Four design rules

R1 (review-interval sizing). Within a declared hold-until-review class — the institutional reading of a sampled-review regulator — a review period exceeding the worst-case exit time renders in-window enforcement futile: the timing certificate (Theorem 3 of Abaee, 2026a) certifies nonviability exactly when T_{obs} exceeds the exit-time value. The scope qualifier is load-bearing: the statement is about the declared class, and the stochastic companion quantifies the contrast — on the audited delayed-regime grid the declared-class value and the unrestricted sequential value coincide at the one-step horizon and differ exactly on the timing cells for later horizons, where an open-loop time-varying sequence keeps both regime branches within $[z_0 - 1, z_0 + 1]$ (Abaee, 2026d, Theorem 2(e) and Remark on class scope; verified by the oscillation identity, check S11, and the class table, check S8). On the audited grid the exit-time value is $\sigma^*(z_0) = z_0 - 1$ exactly (48 cells; Abaee, 2026d, checks S4/S15). As a design rule: size the review period against the worst-case class-scoped exit time, and state the class when invoking it.

R2 (the fibre rule for indicator aggregation). An aggregated indicator can support an exact safe/unsafe verdict on a floor only if its fibres do not merge states requiring incompatible actions: no threshold on a composite index can guarantee a specific coordinate floor in general (Proposition 7 and Corollary 1 of Abaee, 2026a, which apply verbatim to composite indices). The rule has an exact one-dimensional design form — every observation fibre must carry an instrument jointly safe for all its states — and its cost reading is computed on the audited target: the coarsest admissible monitoring costs two cells, and the pairwise version of the rule is necessary but not sufficient off convexity, with an exact three-codex counterinstance (Abaee, 2026f, lineage *From Obstruction to Design*, checks E1–E3). As a design rule: coordinate-wise floor monitoring is the sound default; an aggregate is admissible only with certified compatible fibres.

R3 (bias correction). A known reading bias empties the kernel of an uncorrected certainty-equivalence controller even when the perfect-information kernel is nonempty (Remark 4 of Abaee, 2026a). In the audited

trap the uncorrected reading drives every class law to drift at least $\frac{21}{100}$ — exit — on the whole audited band, while the corrected law is viable (Abaee, 2026f, lineage *Beyond Constant Drift*; Abaee, 2026d, check S1’s two-floor instance for the corresponding chance bound). As a design rule: a biased indicator is admissible only under a correction certified against the model, not merely acknowledged.

R4 (structural-cause forensics). The certificates separate two post-mortem narratives that are otherwise conflated: managerial failure and information-structure impossibility. The common-action obstruction certifies that the reading merged states requiring opposite interventions — collapse guaranteed before any decision — with exact instances throughout the programme (the audited belief $\{(1, 2), (2, 1)\}$: nonviable under the aggregate reading, viable under either coordinate reading; Abaee, 2026d, check S5; the unanimity deadlock of decentralized observation, sixteen coordinated- viable pairs lost under split rights, Abaee, 2026f, lineage *Decentralized Observation*). As a design rule: a collapse audit should test the information structure before the personnel.

Remark 1 (what the rules are). Each rule is a restatement of a theorem at design precision, with its scope and its verified instance attached. None is a heuristic; none extends its source. Where a rule is class-scoped (R1), the scope is part of the rule.

3. Operational versus architectural use

The calculus is not an operational control tool: it does not steer a fishery, a reservoir, or a vehicle, and no part of the programme claims otherwise. Its use is architectural and forensic: (i) *ex-ante design* — proving a proposed monitoring arrangement (aggregation, review period, bias treatment, sensor suite) mathematically unable to support safety, before it is built; (ii) *standard-setting* — converting qualitative objections to composite-index compliance into impossibility theorems with the precision required in regulatory texts; (iii) *forensics* — separating structural from managerial cause after failure; and (iv) *sensor-sufficiency certification* in safety-critical automation — certifying a degraded sensing configuration inadequate against worst-case disturbance, the direction opposite to conventional safety verification, with the programme’s policy-class and hybrid-mode lineages as the natural instruments (Abaee, 2026f, lineages *The Policy-Class Lattice* and *Discrete Regimes, Carefully*).

4. Stakeholder scope

5. Limitations, stated precisely

L1 (model availability). The calculus consumes a model: dynamics, disturbance bounds, floor boundaries, instrument sets. Ecological and institutional systems rarely arrive with calibrated equations and tight bounds. The answer is not to shrink the requirement but to name it: *bounding is the regulatory act*. Declaring the disturbance class and the floor margins is precisely the deliberation the calculus formalizes, and the buffered-viability lineage prices margins into the certificates (Abaee, 2026f, lineage *Buffered Viability*: the obstruction margin, e.g.

$\mu = 2$ on the audited beliefs, erosion $28 \rightarrow 3 \rightarrow 0$). Where no defensible bounds exist, the calculus is silent — by construction, not by oversight.

L2 (dimensionality). Exact recursions are exponential in the worst case ($|\Gamma_{k+1}| \leq |A| |\Gamma_k|^{|Y|}$; Abaee, 2026d, Proposition 1); the audited instances are one- and two-dimensional, and ten-to-thirty-variable ecosystem models are not auditable by the exact route. The continuous-to-finite bridge carries LP-level certificates for polyhedral classes (Abaee, 2026b, solver-certified campaign), the timing-instantiation theorem reduces the timing check to finite linear programs over polytope-declared classes (Abaee, 2026a, Theorem 4), and the finite-horizon completeness lineage gives an exact game algorithm on finite systems (Abaee, 2026f, lineage *Complete on the Finite Horizon*). A medium-dimensional numerical campaign remains deferred in the calculus manuscript and is the main computational gap.

L3 (completeness scope). Completeness of the obstruction calculus holds on finite horizons for finite systems (Abaee, 2026f, lineage *Complete on the Finite Horizon*: soundness, completeness, and counterstrategy trees; the certificate gap is empty there); for continuous blind-window control classes the check remains a template (Open Problem 1 of Abaee, 2026a), and the residual dynamic gap of its §6.5 stands. Statements beyond these scopes are not claimed.

L4 (calibration and case studies). No result in the programme is calibrated against real schedules, stocks, or sensor data. The two-patch, hidden-regime, and trap instances are deliberately small and exact; the real-schedule illustration in the calculus manuscript was marked optional and not taken. Applied calibration is an open, owner-side item.

L5 (software status). The programme’s software is verification-grade: exact rational scripts regenerating every claimed number (sixty check families in the catalogue alone; fifteen in the stochastic lineage), plus a solver-certified LP campaign. There is no user-facing decision-support tool; the distance from a verification script to a tool is an engineering programme of its own and is recorded as such.

L6 (formalizing the intuitive). That aggregation hides trade-offs, that late review arrives too late, that uncorrected bias misleads — all intuitive. The contribution is that these are theorems: in standard-setting and forensic contexts, intuition is not admissible evidence, and an impossibility proof with a verified certificate is. The calculus’ value is not that its conclusions surprise experts; it is that its conclusions bind.

6. The delivered programme against the utility prerequisites

The follow-up programme was drafted against exactly the prerequisites an operational reading would set. Index: continuous-to-finite checkable certificates — delivered

Stakeholder	Reach today
Field operator	none direct; rules R1–R3 enter as institutional
Standards body	R1–R2 as auditable design constraints
Post-mortem auditor	R4 as a structural-cause test
Safety engineer (CPS)	sensor-sufficiency certification (outlook)
Viability/control researchers	the full calculus and programme

Table 1: Who can use the results today, and at which level.

(Abaee, 2026b, exact primal/dual LP witnesses); belief-state value theory — delivered (Abaee, 2026d, fifteen exact check families); monitoring design rules — delivered (Abaee, 2026f, lineage *From Obstruction to Design*: fibre conditions, coarsest monitoring, delay/bias/aggregation rules); diagnostics for restriction attribution — delivered (Abaee, 2026f, lineage *The Policy-Class Lattice*: computed distances to viability); exit-time theory beyond constant drift — delivered (Abaee, 2026f, lineage *Beyond Constant Drift*); uncertainty, regimes, margins, and institutional splits — delivered (Abaee, 2026f, lineages *Hidden Parameters and Learning Deadlines* through *Decentralized Observation*); architecture consolidating the five layers — delivered (Abaee, 2026e). Remaining: user-facing tooling (L5), applied calibration (L4), the deferred numerical campaign (L2), and the open completeness problems (L3).

7. Machine-checked pointers

`paper2_applications_note_v1_pointers.py` probes the cited artifacts (the shipped PDFs of the programme) for each object this statement quotes — the theorems, the verified instances, and the lineage titles — and reports the pointer record. It requires a PDF text extractor (`pymupdf`, stated dependency); all other checks are exact string identities on normalized text. A pointer that fails breaks the statement; the record ships with it.

References

- Abaee, A.: An obstruction calculus for viability under incomplete observation. Preprint. <https://github.com/MIKEAA2020/general-sustainability> (2026a)
- Abaee, A.: Finite nonviability certificates under delayed information. Preprint. <https://github.com/MIKEAA2020/general-sustainability> (2026b)
- Abaee, A.: The viable selector: a unifying framework for obstruction certificates under incomplete observation. Preprint. <https://github.com/MIKEAA2020/general-sustainability> (2026c)
- Abaee, A.: The stochastic selector: exact rational belief-state safety values under partial observation, 2nd edn. Preprint. <https://github.com/MIKEAA2020/general-sustainability> (2026d)
- Abaee, A.: Five layers of obstruction: a successor architecture for viability under incomplete observation. Preprint. <https://github.com/MIKEAA2020/general-sustainability> (2026e)
- Abaee, A.: Generalization catalogue of the obstruction programme, ten lineages: vector floors; finite-horizon completeness; certificate duality; the policy-class lattice; monitoring design; comparison-function exit; hidden parameters and learning deadlines; hybrid modes; buffered viability; decentralized observation. Preprint. <https://github.com/MIKEAA2020/general-sustainability> (2026f)

Declarations

Funding. None. **Competing interests.** None.
Data availability. No external data were used.

Code availability. The pointer-check script `paper2_applications_note_v1_pointers.py` is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex` in the repository); it verifies every pointer of Section 7 against the shipped artifacts. **AI declaration.** GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review within the programme’s verification discipline.